



FORT STREET HIGH SCHOOL

Name: \_\_\_\_\_

Teacher: \_\_\_\_\_

Class: \_\_\_\_\_

## 2018 ASSESSMENT TASK 3 – PRELIMINARY

# Mathematics Extension 1

**Time allowed: 90 minutes**  
(plus 5 minutes reading time)

| Syllabus Outcomes | Assessment Area Description and Marking Guidelines                                                                                                                                                                                                             | Questions |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| P4                | chooses and applies appropriate arithmetic, algebraic, graphical and trigonometric techniques                                                                                                                                                                  | 1, 3, 5   |
| P6<br>PE5<br>PE6  | Understands the concept of a function and the relationship between a function and its graph, determines the derivatives which require the application of more than one rule of differentiation, relates the derivative of a function to the slope of its graph | 4, 2, 5   |
| PE6               | makes comprehensive use of mathematical language, diagrams and notation for communicating in a wide variety of situations                                                                                                                                      | 5         |

### General Instructions:

- Questions 1-5 are to be started in a new booklet
- The marks allocated for each question are indicated
- In Questions 1-5, show relevant mathematical reasoning and/or calculations.
- Marks may be deducted for careless or badly arranged work.
- Board – approved calculators may be used

### Total Marks 58

Attempt Questions 1-5

| Question | Out of   | Marks |
|----------|----------|-------|
| Q1       | /10      |       |
| Q2       | /12      |       |
| Q3       | /11      |       |
| Q4       | /12      |       |
| Q5       | /13      |       |
| Total    | /58      |       |
|          | Percent: |       |



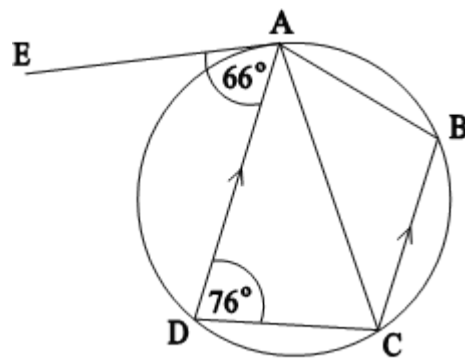
**Question 1 (10 marks)** Use a SEPARATE writing booklet.

- a) Given that  $R(x, y)$  divides the interval joining  $P(-4, 5)$  and  $Q(-1, 9)$  externally in the ratio 3:5, find the coordinates of  $R$ . 2

- b) i) Sketch  $y = |x| - x$  2

- ii) Hence or otherwise, solve  $|x| - x > 2$  1

- c) In the circle shown in the diagram,  $ABCD$  is a cyclic quadrilateral in which  $AD$  is parallel to  $BC$  and  $\angle ADC = 76^\circ$ .  $AE$  is tangent to the circle such that  $\angle EAD = 66^\circ$ .  
**Copy diagram into your solution booklet.**



- i) Find the value of  $\angle DAC$ . Give reasons. 2

- ii) Show that the triangle  $ABC$  is isosceles. 3

**End of question 1**

**Question 2 (12 marks)** Use a SEPARATE writing booklet.

- a) i) Find the equation of the locus of a point  $P(x, y)$  moving so that  $PA$  is perpendicular to  $PB$ , where  $A = (3, 4)$  and  $B = (-1, 2)$ . 2
- ii) Show that the locus is a circle and state its centre and radius. 3
- b) i) State the focal length and the equation of the directrix of a parabola with focus  $(3, 1)$  and vertex  $(5, 1)$ . 2
- ii) State the equation of the parabola. 1
- iii) Find the coordinates of the endpoints of the latus rectum. 2
- c) The point  $P(k, 6)$  lies on the parabola  $x^2 = 16y$ . Find the distance from  $P$  to the focus of the parabola. **Show working out/ reasoning** 2

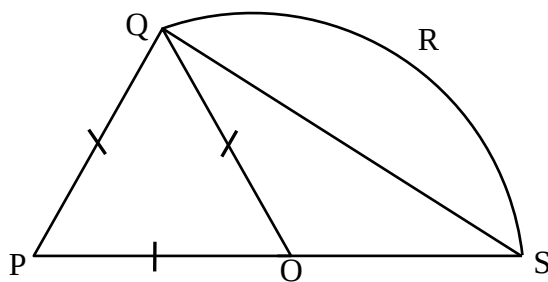
**End of question 2**

**Question 3 (11 marks)** Use a SEPARATE writing booklet.

a) Find the exact value of  $300^\circ$  in radians. 1

b) Solve  $2\sin 2\alpha - 1 = 0$  for  $0 \leq \alpha \leq 2\pi$ . 3

c)  $OPQ$  is an equilateral triangle of side  $12\text{cm}$ .  $PS$  is a straight line.  $QRS$  is an arc of a circle with centre  $O$ .



Giving your answer in exact form find:

i. the perimeter of the region  $PQRSOP$  3

ii. the area of the segment  $SQR$  (i.e. the segment bounded by the chord  $QS$  and the arc  $QRS$ ) 1

d) Express  $2\cos\theta + 2\cos\left(\theta + \frac{\pi}{3}\right)$  in the form  $R\cos(\theta + \alpha)$  where  $R > 0$  and  $0 < \alpha < \frac{\pi}{2}$ .

**Show all working out.** 3

**End of question 3**

**Question 4 (12 marks)** Use a SEPARATE writing booklet.

a) Differentiate the following:

i)  $y = 4x^2 - 5x$  1

ii)  $y = \frac{1}{2x-4}$  1

b) Consider the function  $f(x) = \frac{2x^2}{x^2 + 1}$

i) Find the equation of any horizontal asymptotes. **Show all working out** 2

ii) Using the first derivative, find the coordinates of any turning point(s) and determine their nature. 4

iii) List all the steps for identifying the  $x$  coordinate of any points of inflexion. **Do not perform any calculations.** 1

iv) Sketch the graph of this function, showing all intercepts and asymptotes. 3

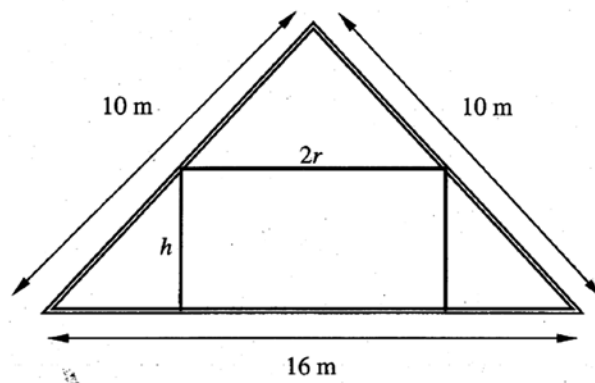
**End of question 4**

**Question 5 (13 marks)** Use a SEPARATE writing booklet.

- a) Draw a sketch of a function that has  $f'(x) > 0$  for  $x < -2$  and  $x > 2$ ,  $f'(x) = 0$  at  $x = -2$  and at  $x = 2$ , and  $f'(x) < 0$  for  $-2 < x < 2$

2

- b) In some rural areas, hot water tanks are installed in the roofs of large homesteads. The diagram below shows the cross-section of a cylindrical tank in such a homestead's roof. The cylindrical tank, with diameter  $2r$  metres and height  $h$  metres, fits exactly into the roof. The cross-section of the roof is an isosceles triangle with dimensions shown.



- i) Show that the height of the roof is 6 metres.

1

- ii) Hence, show that  $h = 6 - \frac{3}{4}r$

2

- iii) Show that the volume of the cylindrical tank can be expressed by

1

$$V = 3\pi r^2 \left( 2 - \frac{r}{4} \right)$$

- iv) Find the value of  $r$  which gives the hot water tank its greatest volume.

3

- c) Find all the values of  $\theta$  for which  $8\sin^2 \theta - 2\cos^2 \theta = \cos 2\theta$

4

**End of Examination**



**2018**

**HIGHER SCHOOL CERTIFICATE PRELIMINARY COURSE**

## **ASSESSMENT TASK 3**

# Mathematics Extension 1

## Solutions

### Question 1 (10 marks)

- a) Given that  $R(x, y)$  divides the interval joining  $P(-4, 5)$  and  $Q(-1, 9)$  externally in the ratio 3:5, find the coordinates of  $R$ .

2

#### **Solution :**

Divided externally  $\Rightarrow$  the ratio is 3:-5

$$\begin{aligned}x &= \frac{mx_2 + nx_1}{m+n} & y &= \frac{my_2 + ny_1}{m+n} \\&= \frac{-5(-4) + 3(-1)}{-5+3} & &= \frac{-5(5) + 3(9)}{-5+3} \\&= -\frac{17}{2} & &= -1\end{aligned}$$

$$\therefore R\left(-\frac{17}{2}, -1\right)$$

#### **Marking guidelines**

- 2 Both  $x$  and  $y$  are correct  
1 1 of either  $x$  or  $y$  correct

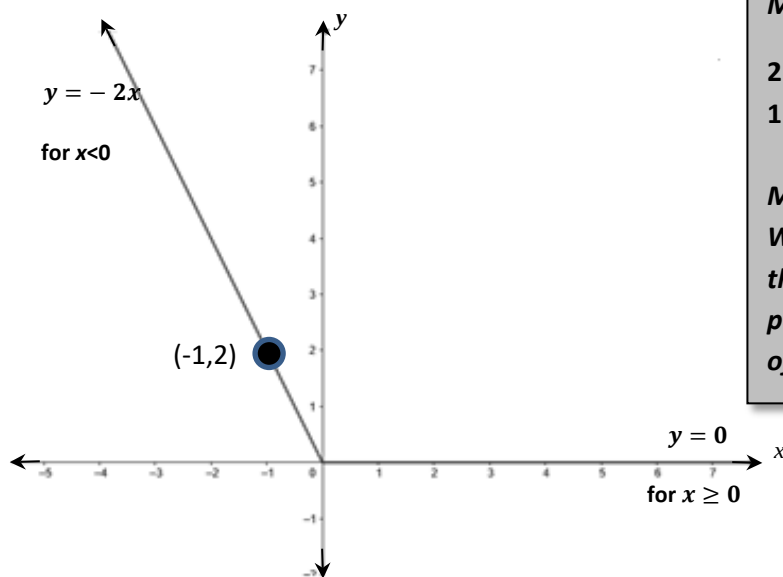
#### **Marker's comments**

*Well done. A few students made arithmetic errors.*

- b) i) Sketch  $y = |x| - x$

2

#### **Solution :**



#### **Marking guidelines**

- 2 Correct sketch  
1 Either ray correctly drawn

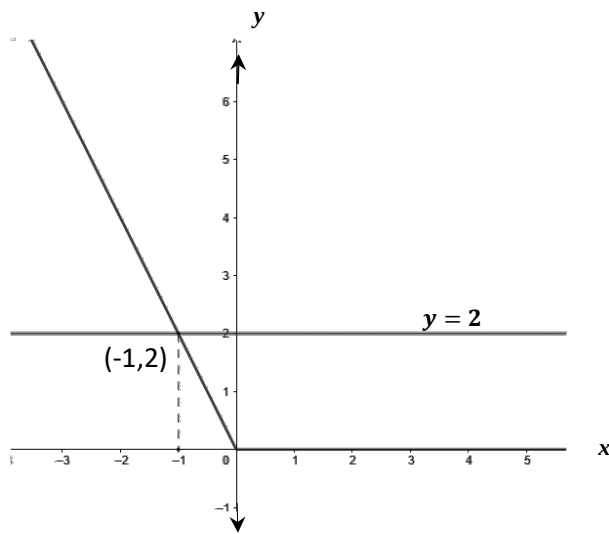
#### **Marker's comments**

*Well done. Some students left out scales on the axes or didn't give the co-ordinates of a point on the ray  $y = -2x$  for  $x < 0$ . At least one of these should be shown.*

ii) Hence or otherwise, solve  $|x| - x > 2$

1

**Solution :**  $x < -1$



**Marking guidelines**

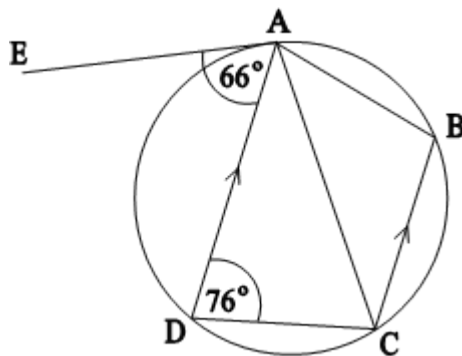
**1** Correct solution

**Marker's comments**

**Well done.**

- c) In the circle shown in the diagram,  $ABCD$  is a cyclic quadrilateral in which  $AD$  is parallel to  $BC$  and  $\angle ADC = 76^\circ$ .  $AE$  is tangent to the circle such that  $\angle EAD = 66^\circ$ .

**Copy diagram into your solution booklet.**



i) Find the value of  $\angle DAC$ . Give reasons.

2

**Solution :**

$\angle ACD = 66^\circ$  (angle between tangent and chord is equal to angle in the alternate segment)

$\angle DAC + 76^\circ + 66^\circ = 180^\circ$  (angle sum of a triangle is  $180^\circ$ )

Hence,  $\angle DAC = 38^\circ$

**Marking guidelines**

**2** Correct value of  $\angle ACD$  with correctly stated reasons.

**1** Either  $\angle ACD$  or  $\angle DAC$  correct

**Marker's comments**

*Many students wrote a poorly worded reason for the "The angle between the tangent and chord, at point of contact, equals the angle in the alternate segment."*

ii) Show that the triangle ABC is isosceles.

3

**Solution :**

$\angle ACB = 38^\circ$  (alternate angles on parallel lines)

$\angle ABC = 180^\circ - 76^\circ = 104^\circ$  (opposite angles of cyclic quadrilateral are supplementary)

$\therefore \angle CAB + 104^\circ + 38^\circ = 180^\circ$  (angle sum of triangle ABC is  $180^\circ$ )

$\therefore \angle CAB = 38^\circ$

$\therefore \angle CAB = \angle ACB = 38^\circ$

$\therefore \triangle ABC$  is isosceles (two angles equal)

**Marking guidelines**

**3** Correct proof

**2** Correctly finding 2 angles with correctly stated reasons.

**1** Correctly finding an angle with a correctly stated reason.

**Marker's comments**

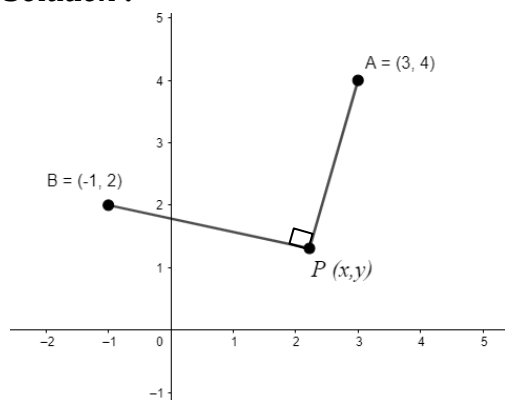
*Well done*

## Question 2 (12 marks)

- a) i) Find the equation of the locus of a point  $P(x, y)$  moving so that  $PA$  is perpendicular to  $PB$ , where  $A = (3, 4)$  and  $B = (-1, 2)$ .

2

**Solution :**



$$\text{Grad. } PA = -\frac{1}{\text{Grad. } PB}$$

$$m_{PA} = \frac{y-4}{x-3} \quad m_{PB} = \frac{y-2}{x+1}$$

Hence,

$$\frac{y-4}{x-3} = -\left(\frac{x+1}{y-2}\right)$$

$$(y-4)(y-2) = -(x+1)(x-3)$$

$$y^2 - 6y + 8 = -(x^2 - 2x - 3)$$

$$\therefore x^2 - 2x + y^2 - 6y + 5 = 0$$

### Marking guidelines

**2** Correct solution

**1** 
$$\frac{y-4}{x-3} = -\left(\frac{x+1}{y-2}\right)$$

### Marker's comments

- Marks awarded for alternate methods – using of Pythagoras Theorem or Circle Geometry Theorems
- Some students recognised it was a circle with diameter AB and found the midpoint to determine the centre of the circle.
- A few students incorrectly used the distance formula or equation of a line
- A few students were unable to correctly recall and apply the gradient formula

- ii) Show that the locus is a circle and state its centre and radius.

3

**Solution :**

$$x^2 - 2x + y^2 - 6y + 5 = 0$$

$$x^2 - 2x + \left(\frac{-2}{2}\right)^2 + y^2 - 6y + \left(\frac{-6}{2}\right)^2 = -5 + \left(\frac{-2}{2}\right)^2 + \left(\frac{-6}{2}\right)^2$$

$$(x-1)^2 + (y-3)^2 = 5$$

Therefore it is a circle since it's in the form

$$(x-h)^2 + (y-k)^2 = r^2. \text{ Centre } (1,3) \text{ and radius } \sqrt{5}.$$

### Marking guidelines

**3** Correct solution

**2** Completing square and writing in the form  $(x-1)^2 + (y-3)^2 = 5$

**1** Completing square, correct radius or correct centre

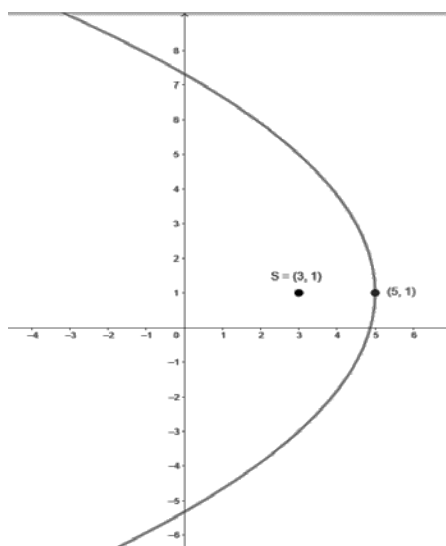
### Marker's comments

- A few students did not attempt the question
- This question uncovered weak algebraic skills some students were unable to complete the square correctly.

- b) i) State the focal length and the equation of the directrix of a parabola with focus (3,1) and vertex (5,1).

2

**Solution :**



Focal Length:  $a = 5 - 3$   
 $= 2$

Equation of directrix:  $x = 5 + 2$   
 $x = 7$

**Marking guidelines**

- 2 Correct focal length and equation of directrix  
 1 Correct focal length or equation of directrix

**Marker's comments**

- Some students did not answer the question fully and left out either the focal length or directrix equation
- Some had incorrect orientation of the parabola
- A few incorrectly gave the equation of the directrix as a point
- Students who used the  $a$  diagram tended to arrive at the correct answer

- ii) State the equation of the parabola.

1

**Solution :**

Parabola is in the form  $(y - k)^2 = -4a(x - h)$

Hence equation is  $(y - 1)^2 = -8(x - 5)$

**Marking guidelines**

- 1 Correct solution

**Marker's comments**

- Not answered very well
- Many students didn't realise it was a sideways parabola which led to problems when attempting to answer part (iii)
- A few swapped the  $x$  and  $y$  coordinates in the equation
- Those that used the correct coordinates for the vertex forgot the negative in front of the 8.

iii) Find the coordinates of the endpoints of the latus rectum.

2

**Solution :**

Latus rectum passes through the focus.

Hence end points are  $(3, y_1)$  and  $(3, y_2)$

When  $x = 3$

$$(y-1)^2 = -8(3-5)$$

$$(y-1)^2 = 16$$

$$y-1 = \pm 4$$

$$\therefore y = 5 \text{ or } y = -3$$

$\therefore$  Coordinates of endpoints are  $(3, 5)$  and  $(3, -3)$

**Marking guidelines**

2 Correct solution

1  $x = 3$

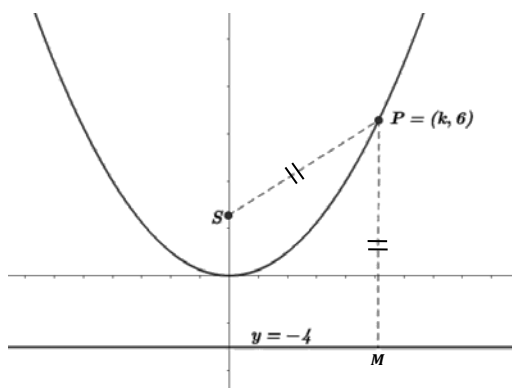
**Marker's comments**

- *Not answered very well – it was apparent that many students didn't fully understand the concept*
- *Weak algebraic skills were apparent with few students able to solve the quadratic once  $x = 3$  was substituted.*
- *Because of the error in part (ii) some students were trying to find the square root of a negative number*

c) The point  $P(k, 6)$  lies on the parabola  $x^2 = 16y$ . Find the distance from  $P$  to the focus of the parabola. **Show working out/ reasoning**

2

**Solution :**



$x^2 = 16y$ , hence focal length,  $a = 4$ .

Therefore, equation of directrix is  $y = -4$ .

Perpendicular distance from  $P(k, 6)$  to the directrix  
 $= 6 + 4$   
 $= 10$

Since distance from any point  $P(x, y)$  on a parabola to the directrix is equal to distance from  $P(x, y)$  to the focus then distance from  $P(k, 6)$  to the focus  $= 10$

**Marking guidelines**

2 Correct solution

1  $a = 4$  or equation of directrix  $y = -4$

**Marker's comments**

- *Not answered very well*
- *Almost all students did not realise they could use the geometric definition of the parabola (distance from a point to the directrix is equal to the distance from the point to the focus). However, marks were awarded if the correct answer was achieved using the distance formula.*
- *When labelling coordinates of the focus, a few were confusing  $x$  and  $y$*

**Question 3 (11 marks)**

- a) Find the exact value of  $300^\circ$  in radians.

1

**Solution :**

$$\begin{aligned} 1^\circ &= \frac{\pi}{180} \\ 300^\circ &= \frac{300\pi}{180} \\ &= \frac{5\pi}{3} \text{ radians} \end{aligned}$$

**Marking guidelines**

**1** Correct solution

**Marker's comments**

**Generally well done**

- b) Solve  $2\sin 2\alpha - 1 = 0$  for  $0 \leq \alpha \leq 2\pi$ .

3

**Solution :**

$$2\sin 2\alpha - 1 = 0 \quad 0 \leq 2\alpha \leq 4\pi$$

$$\sin 2\alpha = \frac{1}{2} \quad \text{related angle} = \frac{\pi}{6}$$

Hence,

$$2\alpha = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{13\pi}{6}, \frac{17\pi}{6}$$

$$\therefore \alpha = \frac{\pi}{12}, \frac{5\pi}{12}, \frac{13\pi}{12}, \frac{17\pi}{12}$$

**Marking guidelines**

**3** Correct solution

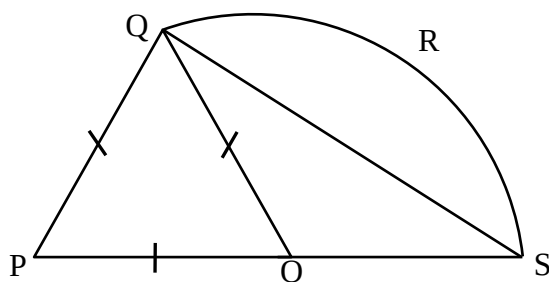
**2** Related angle and 2 correct solutions

**1** Related angle or change of domain

**Marker's comments**

**Many students didn't explicitly state the change of domain. Most of those students only gave 2 out of 4 correct solutions.**

- c)  $OPQ$  is an equilateral triangle of side  $12\text{cm}$ .  $PS$  is a straight line.  $QRS$  is an arc of a circle with centre  $O$ .



Giving your answer in exact form find:

- i. the perimeter of the region  $PQRSOP$

3

**Solution :**

$$\angle QOP = \frac{\pi}{3}$$

$$\angle QOS = \pi - \frac{\pi}{3} = \frac{2\pi}{3}$$

$$\text{Length of arc } QRS = r\theta$$

$$= 12 \times \frac{2\pi}{3}$$

$$= 8\pi$$

Perimeter

$$P = 3 \times 12 + 8\pi$$

$$\therefore P = 36 + 8\pi \text{ cm}$$

**Marking guidelines**

3 Correct solution

2  $\angle QOS$  and length of  $\text{arc } QRS$

1  $\angle QOS$  or length of  $\text{arc } QRS$

**Marker's comments**

*Mostly well done, but quite a few students didn't leave in exact form.*

- ii. the area of the segment  $SQR$

1

**Solution :**

$$A = \frac{1}{2} r^2 (\theta - \sin \theta)$$

$$= \frac{1}{2} \times 12^2 \left( \frac{2\pi}{3} - \sin \frac{2\pi}{3} \right)$$

$$= 72 \times \left( \frac{2\pi}{3} - \frac{\sqrt{3}}{2} \right)$$

$$\therefore A = 12(4\pi - 3\sqrt{3}) \text{ cm}^2$$

**Marking guidelines**

1 Correct solution

**Marker's comments**

*Many students didn't refer to the reference sheet which could have saved some time.*

*Quite a few students didn't leave in exact form.*

- d) Express  $2 \cos \theta + 2 \cos \left( \theta + \frac{\pi}{3} \right)$  in the form  $R \cos(\theta + \alpha)$  where  $R > 0$  and  $0 < \alpha < \frac{\pi}{2}$ . 3

**Solution :**

$$\begin{aligned}
 & 2 \cos \theta + 2 \left( \cos \theta + \frac{\pi}{3} \right) \\
 &= 2 \cos \theta + 2 \left( \cos \theta \cos \frac{\pi}{3} - \sin \theta \sin \frac{\pi}{3} \right) \\
 &= 2 \cos \theta + 2 \left( \frac{\cos \theta}{2} - \frac{\sqrt{3} \sin \theta}{2} \right) \\
 &= 2 \cos \theta + \cos \theta - \sqrt{3} \sin \theta \\
 &= 3 \cos \theta - \sqrt{3} \sin \theta
 \end{aligned}$$

Hence,

$$\begin{aligned}
 3 \cos \theta - \sqrt{3} \sin \theta &= R \cos(\theta + \alpha) \\
 3 \cos \theta - \sqrt{3} \sin \theta &= R \cos \theta \cos \alpha - R \sin \theta \sin \alpha \\
 R \cos \alpha &= 3 \dots\dots (1) \\
 R \sin \alpha &= \sqrt{3} \dots\dots(2) \\
 (1)^2 + (2)^2
 \end{aligned}$$

$$\begin{aligned}
 R^2 (\cos^2 \alpha + \sin^2 \alpha) &= 9 + 3 \\
 \therefore R &= 2\sqrt{3}
 \end{aligned}$$

$$2\sqrt{3} \cos \alpha = 3 \qquad 2\sqrt{3} \sin \alpha = \sqrt{3}$$

$$\cos \alpha = \frac{\sqrt{3}}{2} \qquad \sin \alpha = \frac{1}{2}$$

$$\therefore \alpha = \frac{\pi}{6}$$

$$\therefore 2 \cos \theta + 2 \cos \left( \theta + \frac{\pi}{3} \right) = 2\sqrt{3} \cos \left( \theta + \frac{\pi}{6} \right)$$

**Marking guidelines**

- |          |                                                                                                                               |
|----------|-------------------------------------------------------------------------------------------------------------------------------|
| <b>3</b> | <b>Reduction to <math>3 \cos \theta - \sqrt{3} \sin \theta</math>, correct <math>R</math> and correct <math>\alpha</math></b> |
| <b>2</b> | <b>1 mark deducted for 1 error in reduction, <math>R</math> or <math>\alpha</math></b>                                        |
| <b>1</b> | <b>2 marks deducted for 2 errors</b>                                                                                          |

**Marker's comments**

**Many students failed to change the expression to the form  $a \cos \theta - b \sin \theta$  before trying to determine  $R$  and  $\alpha$**   
**Those students who successfully simplified the expression were mostly able to complete a full solution.**

#### Question 4 (12 marks)

a) Differentiate the following:

1

i)  $y = 4x^2 - 5x$

**Solution :**

$$\frac{dy}{dx} = 8x - 5$$

**Marking guidelines**

**1** Correct answer

**Marker's comments**

*Well done.*

ii)  $y = \frac{1}{2x-4}$

1

**Solution :**

$$y = (2x - 4)^{-1}$$

$$\frac{dy}{dx} = -1 \times 2(2x - 4)^{-2}$$

$$\therefore \frac{dy}{dx} = \frac{-2}{(2x - 4)^2}$$

**Marking guidelines**

**1** Correct answer

**Marker's comments**

*Generally well done but there were some errors with indices.*

b) Consider the function  $f(x) = \frac{2x^2}{x^2 + 1}$

i) Find the equation of any horizontal asymptotes. **Show all working out**

2

**Solution :**

$$\begin{aligned} \lim_{x \rightarrow \infty} \frac{2x^2}{x^2 + 1} &= \lim_{x \rightarrow \infty} \frac{\frac{2x^2}{x^2}}{\frac{x^2}{x^2} + \frac{1}{x^2}} \\ &= \lim_{x \rightarrow \infty} \frac{2}{1 + \frac{1}{x^2}}, \quad \lim_{x \rightarrow \infty} \frac{1}{x^2} = 0 \end{aligned}$$

Hence,

$$\lim_{x \rightarrow \infty} \frac{2}{1 + \frac{1}{x^2}} \Rightarrow \frac{2}{1 + 0}$$

$\therefore$  Horizontal asymptote is  $y = 2$

**Marking guidelines**

**2** Division by  $x^2$ ,  $\lim_{x \rightarrow \infty} \frac{1}{x^2} = 0$  and  $y = 2$

**1** Division by  $x^2$  or  $\lim_{x \rightarrow \infty} \frac{1}{x^2} = 0$  or  $y = 2$

**Marker's comments**

*Students had a reasonable idea of concept but couldn't set it out properly.*

*Some students did not answer the question!*

*i.e. concluding statement.*

*HA and VA are NOT acceptable abbreviations.*

*Note: cannot divide by  $\infty$*

- ii) Using the first derivative, find the coordinates of any turning point(s) and determine their nature.

4

**Solution :**

$$f(x) = \frac{2x^2}{x^2 + 1}$$

$$f'(x) = \frac{vu' - uv'}{v^2} \quad u = 2x^2 \quad v = x^2 + 1$$

$$u' = 4x \quad v' = 2x$$

$$f'(x) = \frac{4x(x^2 + 1) - 2x^2(2x)}{(x^2 + 1)^2}$$

$$\therefore f'(x) = \frac{4x}{(x^2 + 1)^2}$$

Stationary Point at  $f'(x) = 0$

$$\frac{4x}{(x^2 + 1)^2} = 0$$

$$4x = 0$$

$$x = 0$$

$\therefore$  Coordinates of S.P = (0,0)

|         |    |   |   |
|---------|----|---|---|
| $x$     | -1 | 0 | 1 |
| $f'(x)$ | -1 | 0 | 1 |



Therefore, there is a minimum turning point at (0,0)

- iii) List all the steps for identifying the  $x$  coordinate of any points of inflexion. **Do not perform any calculations.**

1

**Solution :**

1. Find second derivative
2. Set second derivative to 0 and solve for  $x$
3. Setup sign table, find values of  $f''(x)$  on either side of  $x$  from step 2
4. If there is a change in sign of  $f''(x)$  on either side of  $x$  then there is a change in concavity and a point of inflexion exists.

#### Marking guidelines

**4** 1<sup>st</sup> derivative using quotient rule  
S.P. at (0, 0)

Sign table

Minimum/ concave up at (0, 0)

**3** 1 error

**2** 2 errors

**1** 3 errors

#### Marker's comments

*Some students confused first derivative with first principles which wasted time and often led to errors.*

*Students MUST state for stationary points  $f'(x) = 0$*

*Many students ignored the instruction and used the second derivative to determine the nature.*

#### Marking guidelines

**1** ALL steps listed

#### Marker's comments

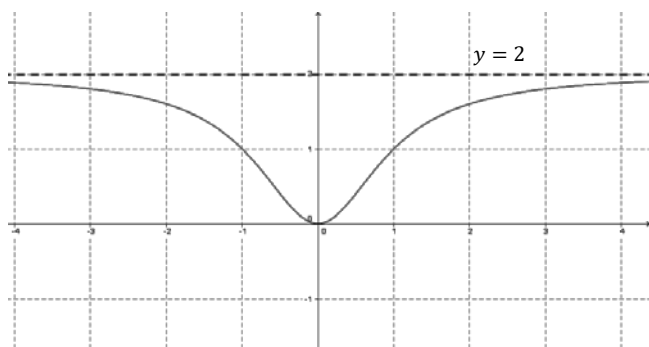
*Poorly done. Some students ignored instructions and performed calculations- no marks.*

*There seems to be a lot of confusion between inflexion points and a horizontal point of inflexion!*

iv) Sketch the graph of this function, showing all intercepts and asymptotes.

3

**Solution :**



**Marking guidelines**

- |          |                                                 |
|----------|-------------------------------------------------|
| <b>3</b> | <b>Correct curve to asymptote relationship.</b> |
|          | <b>Turning point at (0,0)</b>                   |
|          | <b>Shape/symmetry of curve</b>                  |
| <b>2</b> | <b>1 feature of curve incorrect</b>             |
| <b>1</b> | <b>2 features of curve incorrect</b>            |

**Marker's comments**

*A sketch requires a ruler to be used, axes to be labelled with reasonable relative scale and should be a suitable size.*

*Many had a 'V' instead of a minimum turning point.*

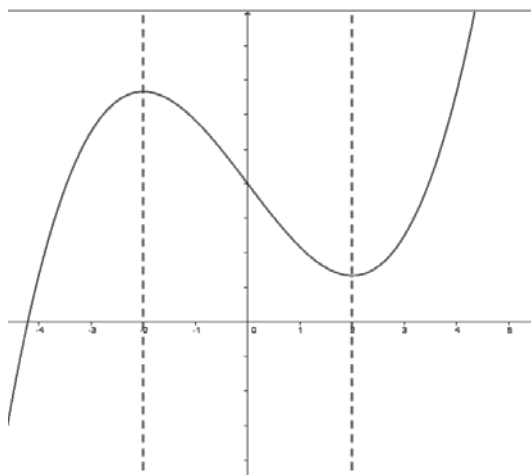
*More care required with shape and the curve as it approaches the asymptote.*

### Question 5 (13 marks)

- a) Draw a sketch of a function that has  $f'(x) > 0$  for  $x < -2$  and  $x > 2$ ,  $f'(x) = 0$  at  $x = -2$  and at  $x = 2$ , and  $f'(x) < 0$  for  $-2 < x < 2$

2

**Solution :**



Turning points at  $x = 2$  and  $x = -2$

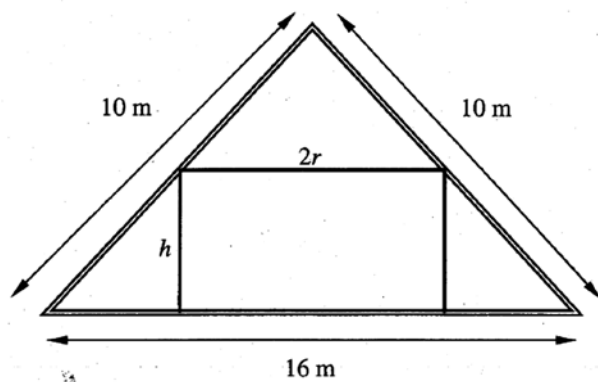
#### Marking guidelines

- 2 Correct shape and turning points at  $x = 2$  and  $x = -2$   
1 Correct shape or turning points

#### Marker's comments

Students should label their axes and graph.

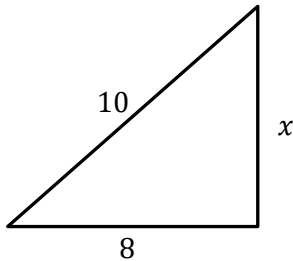
- b) In some rural areas, hot water tanks are installed in the roofs of large homesteads. The diagram below shows the cross-section of a cylindrical tank in such a homestead's roof. The cylindrical tank, with diameter  $2r$  metres and height  $h$  metres, fits exactly into the roof. The cross-section of the roof is an isosceles triangle with dimensions shown.



- i) Show that the height of the roof is 6 metres.

1

**Solution :**



$$\begin{aligned}x^2 &= 10^2 - 8^2 \\&= 36 \\ \therefore x &= 6m \text{ as req'd}\end{aligned}$$

**Marking guidelines**

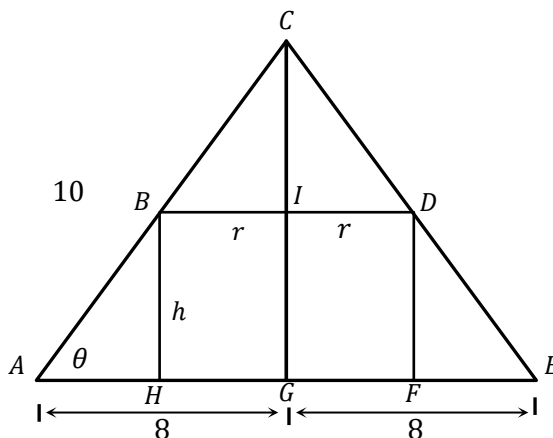
**1** Correct answer

**Marker's comments**

Generally well done.

- ii) Hence, show that  $h = 6 - \frac{3}{4}r$

2



Let  $\angle CAE = \theta$

$$\text{In } \triangle ACG \tan \theta = \frac{6}{8} = \frac{3}{4}$$

$\triangle ACG \parallel \triangle ABH$  hence,

$$\tan \theta = \frac{h}{8-r} \text{ in } \triangle ABH$$

Therefore,

$$\frac{h}{8-r} = \frac{3}{4}$$

$$h = \frac{3}{4}(8-r) = \frac{3}{4} \times 8 - \frac{3}{4}r$$

$$\therefore h = 6 - \frac{3}{4}r \text{ as req'd}$$

**Marking guidelines**

**2** Identifying similar triangles, equating ratios and finding  $h$  in required form

**1** Identifying similar triangles or equating ratios

**Marker's comments**

Poorly answered. Students should think similarity when seeing 'triangles within triangles.'

Some clever students equated areas to find  $h$ .

iii) Show that the volume of the cylindrical tank can be expressed by

1

$$V = 3\pi r^2 \left( 2 - \frac{r}{4} \right)$$

**Solution :**

$$\begin{aligned} V &= \pi r^2 h \\ &= \pi r^2 \left( 6 - \frac{3}{4}r \right) \quad \text{from part (ii)} \\ &= 6\pi r^2 - \frac{3}{4}\pi r^3 \\ &= 3\pi r^2 \left( 2 - \frac{r}{4} \right) \quad \text{as req'd} \end{aligned}$$

**Marking guidelines**

**1 Correct answer**

**Marker's comments**

Generally well answered.

iv) Find the value of  $r$  which gives the hot water tank its greatest volume.

3

**Solution :**

$$\begin{aligned} V &= 6\pi r^2 - \frac{3}{4}\pi r^3 \\ \frac{dV}{dr} &= 12\pi r - \frac{9\pi r^2}{4} \\ \text{Stationary point at } \frac{dV}{dr} &= 0 \\ 12\pi r - \frac{9\pi r^2}{4} &= 0 \\ r \left( 12\pi - \frac{9\pi r}{4} \right) &= 0 \\ r = 0 \quad \text{or} \quad 12\pi - \frac{9\pi r}{4} &= 0 \\ \therefore r &= \frac{16}{3} = 5\frac{1}{3} \text{ m} \\ \frac{d^2V}{dr^2} &= 12\pi - \frac{9\pi r}{2} \\ \text{When } r &= \frac{16}{3} \\ \frac{d^2V}{dr^2} &= 12\pi - \frac{9\pi \times 16}{6} \\ &= -12\pi \\ &< 0 \quad \therefore \text{maximum} \\ \therefore \text{tank radius of } 5\frac{1}{3} \text{ m} &\text{ will give maximum volume} \end{aligned}$$

**Marking guidelines**

**3 Correct with proof that  $r$  is a maximum using the test for max.**

**2 Correctly solved**  $\frac{dV}{dr} = 0$

**1 Correctly found**  $\frac{dV}{dr}$

**Marker's comments**

Many students should practise differentiating.

Clearer setting-out would help minimise careless errors.

Students should be aware that they must demonstrate their value is a maximum by using

either  $\frac{d^2V}{dr^2}$  or  $\frac{dV}{dr}$  sign table.

Students should check the reasonableness of their answers. For example many students had  $r = 16$  giving the maximum value; however looking at the diagram  $r < 8$ .

c) Find all the values of  $\theta$  for which  $8\sin^2 \theta - 2\cos^2 \theta = \cos 2\theta$

4

**Solution :**

$$\begin{aligned} 8\sin^2 \theta - 2\cos^2 \theta &= \cos 2\theta \\ 8(1 - \cos^2 \theta) - 2\cos^2 \theta &= 2\cos^2 \theta - 1 \\ 8 - 8\cos^2 \theta - 2\cos^2 \theta &= 2\cos^2 \theta - 1 \\ 12\cos^2 \theta - 9 &= 0 \\ \cos^2 \theta &= \frac{3}{4} \\ \cos \theta &= \pm \frac{\sqrt{3}}{2} \\ \text{Related acute angle } \alpha &= \frac{\pi}{6} \end{aligned}$$

General solution:

$$\begin{aligned} \theta &= 2n\pi \pm \frac{\pi}{6} \quad \text{and} \\ \theta &= 2n\pi \pm \frac{5\pi}{6} \end{aligned}$$

This can be simplified to

$$\theta = n\pi \pm \frac{\pi}{6}, \text{ where } n \text{ is an integer.}$$

*Alternate solution:*

$$\begin{aligned} 8\sin^2 \theta - 2\cos^2 \theta &= \cos 2\theta \\ 8\sin^2 \theta - 2(1 - \sin^2 \theta) &= 1 - 2\sin^2 \theta \\ 8\sin^2 \theta - 2 + 2\sin^2 \theta &= 1 - 2\sin^2 \theta \\ 12\sin^2 \theta - 3 &= 0 \\ \sin^2 \theta &= \frac{1}{4} \\ \sin \theta &= \pm \frac{1}{2} \\ \text{Related acute angle } \alpha &= \frac{\pi}{6} \end{aligned}$$

General solution:

$$\theta = n\pi + (-1)^n \frac{\pi}{6} \quad \text{and} \quad \theta = n\pi - (-1)^n \frac{\pi}{6}$$

$$\theta = n\pi \pm \frac{\pi}{6}, \text{ where } n \text{ is an integer.}$$

#### **Marking guidelines**

- 4** Finding the simplified expression for  $\theta$  or Finding two general solutions.
- 3** Partially correct general solution or General solutions using degrees.
- 2** Solving for  $\cos \theta$  or  $\sin \theta$  or  $\tan \theta$   
e.g.  $\cos \theta = \pm \frac{\sqrt{3}}{2}$
- 1** Simplifying the equation in terms of one ratio.

#### **Marker's comments**

Very poorly answered.

Many students did not know the trigonometric identities and consequently were unable to simplify the equation.

When solving for  $\cos \theta$  many students did not take the  $\pm$  of the root.

Many students incorrectly answered in degrees (often mixing their answer with radians).

Many students restricted their domain to  $0 < \theta < 2\pi$  and do not realise trig equation have infinite solutions.

Many students missed out on the second solution i.e.  $\theta = 2n\pi \pm \frac{5\pi}{6}$ .

Many students did not explain their pattern by saying "where  $n$  is an integer".